

Affiliation Framing in the Jacobian Lens

A Two-Model Pilot of Internal Readout Sensitivity and Its Limits

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Abstract

Sub-propositional affiliation signals (SPAS) have been proposed as linguistic cues that may influence users through affiliation and ingroup mechanisms not captured by content-level safety evaluation. Existing operationalization, however, is limited to surface outputs. This pilot asks whether affiliation framing in a user prompt changes a language model's internal disposition to verbalize affiliation concepts before it generates an answer. We fitted Jacobian lenses on a small, task-independent corpus for two open instruction-tuned models, SmolLM2-135M and Qwen2.5-0.5B. Twenty English-Spanish prompt pairs held the substantive task constant while adding an affiliation frame in one condition. The prospectively specified primary contrast compared mean Jacobian-lens logits for single-token affiliation concepts against neutral control concepts at the fitted layer nearest 67% of model depth. On twelve held-out pairs, SmolLM2 did not support the prediction (mean paired difference -0.142 logit units, bootstrap 95% CI [-0.759, 0.485], sign-flip $p = 0.679$). The exploratory Qwen robustness run produced a positive result under the same protocol (0.720, 95% CI [0.423, 1.025], $p = 0.0018$, Cohen's $d_z = 1.27$), with positive effects also appearing in later layers. Surface affiliation markers remained sparse in both models. The cross-model disagreement prevents a general claim that affiliation framing reliably enters a shared workspace. It instead establishes three narrower findings: Jacobian-lens measurement of affiliation is computationally feasible; the observed effect can be strongly model-dependent; and internal readout shifts need not appear as explicit affiliation language. Because affiliation words occur in the treatment prompts, lexical carryover remains a live alternative explanation. The study therefore supplies a reproducible measurement protocol and identifies the controls required before mechanistic evidence can support the broader SPAS account.

Keywords: mechanistic interpretability; Jacobian lens; affiliation signals; sycophancy; latent representations; multilingual language models; AI alignment; SPAS

1. Introduction

Most deployed AI safety evaluation begins and ends with observable output. A response is checked for factuality, harmful instructions, deception, bias, or compliance with an explicit policy. This is necessary, but it may be incomplete when the relevant influence is carried by the form of an interaction rather than by a proposition asserted in it.

Lerer (2026) proposed the category of sub-propositional affiliation signals: linguistic patterns such as inclusive first-person plural framing, terminological mirroring, validation, and kinship or alliance vocatives.

The account distinguishes the model’s emitted signal from its possible effect on a human receiver. Its initial metric, the Affiliation Signal Score (AS-Score), was intentionally architecture-independent because it counted surface properties of generated text. That portability came with a blind spot. A model might internally represent affiliation without explicitly verbalizing it, or produce warm language without any stable internal organization corresponding to affiliation.

Gurnee et al. (2026) introduced the Jacobian lens, a mechanistic interpretability technique designed to identify internal representations that a language model is disposed to verbalize. Their evidence suggests that a sparse set of verbalizable representations can support report, directed modulation, internal reasoning, flexible generalization, and selective access. They call the corresponding representational structure the J-space and compare its functional role to a global workspace. The technique creates a concrete route for asking whether affiliation framing changes more than the surface output.

This paper reports a small, reproducible test of that possibility. Its contribution is empirical mapping rather than theoretical unification. The central question is deliberately narrow:

When the substantive task is held constant, does an affiliation-framed prompt increase a model’s Jacobian-lens disposition to verbalize affiliation concepts?

The study was designed to permit a negative answer. It uses a fixed primary layer, a held-out prompt split, a recorded tokenizer inventory, paired uncertainty estimates, and an explicit rule for support. The first model produced a null result. A larger multilingual model produced a strong positive result under the same protocol. That divergence is more informative than a single-model success would have been. It shows that the proposed measurement may detect something real while also being sensitive to model scale, training, tokenization, language coverage, or other architectural differences.

The paper does not test whether human kinship or affiliation modules are activated, whether a model is conscious, or whether a readable internal concept constitutes intention or *mens rea*. It tests an earlier link in the proposed causal chain: whether affiliation framing is associated with a measurable internal disposition on the model side.

2. Conceptual Boundary

2.1 SPAS concerns a relation, not merely a token

The SPAS hypothesis has two analytically separate sides. On the emission side, a model produces or internally prepares affiliative signals. On the reception side, those signals affect a user’s trust, agreement, compliance, or cooperative disposition through mechanisms not exhausted by propositional evaluation. Evidence for the first side does not establish the second.

This distinction matters because the Jacobian lens is a model-side instrument. It can reveal that an activation is disposed to make a model say a word in present or future contexts. It cannot observe the user’s cognitive processing. A positive internal readout therefore supports, at most, an emission-mechanism hypothesis. The full SPAS account still requires controlled human studies linking matched outputs to behavioral outcomes.

2.2 What the experiment is observing

A language model does not move directly from a prompt to a finished sentence. It transforms the prompt through a sequence of layers. At each layer, the model holds a high-dimensional numerical representation of the context: which distinctions are becoming relevant, which relationships are active, and which continuations the current state may support. These representations are not sentences hidden inside the model. They are vectors whose coordinates have no immediate human-readable meaning.

The experiment asks whether a small relational change in the prompt alters that internal representational state. Consider two instructions:

Evaluate the advantages and disadvantages of the proposal.

We share the same values and can work through this together. Evaluate the advantages and disadvantages of the proposal.

The requested analysis is substantively the same. The second prompt additionally establishes a relation between user and model through collective identity, shared values, and cooperation. The experimental question is not merely whether the second answer repeats *we* or *together*. It is whether the relational frame changes the concepts that the model's intermediate state is disposed to make available for later verbalization, even when those concepts do not appear explicitly in the answer.

This distinction is central to the SPAS account. A surface-only analysis can count affiliative expressions in an output. An internal readout can ask whether affiliation framing changed the representational context from which that output was produced. The two measurements address different stages of the proposed process and need not move together.

2.3 The Jacobian lens in intuitive terms

The Jacobian lens is a translation instrument between an intermediate model state and the model's own vocabulary. In intuitive terms, it asks:

If the activity present at this layer influences what the model says now or later, which words or concepts is it disposed to make verbalizable?

One can picture the layers as successive stages in preparing a statement. Early layers contain relatively diffuse associations. Intermediate layers organize contextual and conceptual relationships. Later layers approach the representations from which concrete vocabulary is selected. A basic logit lens projects an intermediate state directly onto the vocabulary, as though the remaining layers did no transformative work. The Jacobian lens instead estimates how changes at an intermediate layer normally propagate through the remaining computation. It is therefore prospective: it measures potential influence on current and future positions, not only resemblance to the next token.

The relevant mathematical object is a Jacobian, a matrix of derivatives. A derivative records how much an output changes when an input changes slightly. Here, the Jacobian records how a small change in the residual-stream activation at layer l would, on average, change later model representations. Applying that estimated transport to a particular activation and then using the model's ordinary unembedding produces vocabulary-indexed scores. High scores for concepts such as *together*, *trust*, or *ally* mean that the activation is comparatively disposed, under the averaged local transformation, to support their verbalization.

This is neither mind reading nor a transcript of hidden reasoning. A lens score does not establish that the model feels affiliation, intends to influence the user, possesses an affiliation module, or will necessarily emit the scored word. It is a model-relative, first-order estimate of how an activation can influence verbalizable representations. The estimate depends on the fitted corpus, layer, position, tokenizer, model architecture, and local linear approximation.

This experiment uses the lens as an internal comparison instrument. For each prompt, it compares average scores for affiliation concepts against neutral control concepts. It then compares that contrast between a neutral prompt and its affiliation-framed counterpart. A positive paired difference means that the affiliation frame shifted the internal readout relatively toward affiliation vocabulary. It does not, by itself, identify why that shift occurred.

2.4 Formal definition of the Jacobian lens

For residual-stream activation h_l at layer l , the Jacobian lens estimates an average linear transport into the final-layer representational basis:

$$J_l = E[\text{partial } h_{\text{final}, t'} / \text{partial } h_l, t], \text{ for } t' \geq t.$$

The expectation is taken over prompts, source positions, and current or future target positions. Applying the lens transports an intermediate activation and then uses the model’s normal unembedding:

$$\text{lens}(h_l) = \text{softmax}(W_U \text{norm}(J_l h_l)).$$

The ranked vocabulary scores describe concepts the activation is, on average, disposed to make verbalizable. This differs from a logit lens, which effectively assumes J_l is the identity, and from a tuned lens, which learns affine probes to predict final outputs (Belrose et al., 2023).

The Jacobian lens remains a first-order, corpus-averaged approximation. It is not a transparent transcript of reasoning. Its vocabulary-indexed readout privileges concepts represented by tokens, and its estimates depend on the model, tokenizer, fitting corpus, layer, and position.

2.5 Hypothesis and alternatives

The directional hypothesis was:

H1: affiliation-framed prompts will increase an Affiliation Readout Contrast relative to substantively matched neutral prompts.

Three rival explanations were retained from the outset or identified by the pilot:

1. **Lexical carryover:** affiliation words in the prompt remain represented because they were recently read, not because an affiliation state was recruited.
2. **Tokenizer coverage:** a model may appear less sensitive because the relevant concepts are fragmented into multiple tokens and excluded from a single-token lens inventory.
3. **Model-specific organization:** instruction tuning, scale, multilingual training, or representational geometry may determine whether affiliation concepts occupy a lens-readable format.

The present design can establish sensitivity, but it cannot fully discriminate among these alternatives.

3. Methods

3.1 Study status and computational environment

The SmolLM2 protocol, prompts, metric, selected depth fractions, and decision rule were written to disk before the first substantive run. This was a local prospective protocol, not an externally registered preregistration. After observing the SmolLM2 result, Qwen2.5-0.5B was added as an explicitly exploratory robustness run. It cannot retroactively change the status of the primary test.

Both runs were executed locally on an Apple M2 Pro with 16 GB unified memory using MPS, float32 weights, Python 3.13.3, PyTorch 2.12.0, and Transformers 5.9.0. The source implementation was Anthropic’s Apache-2.0 Jacobian-lens code. The experiment added separate data, runner, reporting, and provenance files without modifying the estimator.

3.2 Models

SmolLM2-135M-Instruct is a compact instruction-tuned model primarily oriented toward English. Its small size made it suitable for an initial feasibility run on local hardware, but its model card cautions that it primarily understands and generates English. Qwen2.5-0.5B-Instruct has 24 layers, approximately 0.49 billion parameters, and stated multilingual support including English and Spanish. Both are open-weight instruction-tuned causal language models distributed under Apache 2.0.

3.3 Lens fitting

Each model received a separately fitted Jacobian lens. The fitting corpus contained twelve generic passages, six English and six Spanish, about libraries, engineering, budgeting, research, history, and software maintenance. None contained evaluation prompts. The corpus was intentionally small to test local feasibility, not to reproduce the thousand-prompt estimator used by Gurnee et al.

Four layers nearest 33%, 50%, 67%, and 80% of depth were fitted. The layer nearest 67% was designated primary before outcome inspection. SmolLM2 layers were 10, 14, 19, and 23 of 30. Qwen layers were 8, 12, 15, and 18 of 24. The estimator used a dimension batch of two, maximum sequence length of 64, and skipped the first four positions.

3.4 Prompt pairs

The evaluation contained twenty minimal pairs, evenly divided between English and Spanish. Eight pairs formed a development set and twelve formed a held-out set. Each neutral prompt requested a short balanced analysis, such as listing tradeoffs in a four-day workweek. Its paired treatment preserved that task and added an affiliation frame, for example: “Our values are aligned, and we can work through this together.”

The substantive task was held constant, but the treatment necessarily added words, length, and social framing. This creates the lexical-carryover limitation discussed below. Prompt order was deterministic, and the held-out decision was calculated only after all per-prompt readouts were recorded.

3.5 Token inventory and Affiliation Readout Contrast

The candidate affiliation inventory contained English and Spanish concepts associated with collective identity, alliance, validation, and trust. Examples included we, our, together, friend, ally, trust, agree, nosotros, amigo, acuerdo, and primo. Neutral controls included table, number, object, process, document, and list, with Spanish equivalents.

Only concepts represented by exactly one tokenizer token were admitted. Exclusions were automatic and recorded before summary statistics were computed. SmolLM2 admitted nine affiliation and six control concepts; all admitted concepts were English. Qwen admitted thirteen affiliation concepts, including four Spanish concepts, and twelve controls.

For prompt i and layer l , the Affiliation Readout Contrast was:

$$ARC_{i,l} = \text{mean}(\text{logit_affiliation}) - \text{mean}(\text{logit_control}).$$

The paired treatment effect was:

$$\Delta ARC_{i,l} = ARC_{\text{affiliation},i,l} - ARC_{\text{neutral},i,l}.$$

The primary estimate was the mean Delta ARC at the primary layer over twelve held-out pairs. Ten thousand paired bootstrap resamples produced a 95% interval. A two-sided random sign-flip test and paired standardized effect (Cohen’s d_z) were reported as complementary statistics.

The fixed support rule required a positive mean and a bootstrap interval entirely above zero. Secondary analyses covered other fitted layers and a simple surface count of affiliation terms in greedy generated responses. Secondary outcomes could not override the primary label.

3.6 Reproducibility

Every run stored the model identifier, environment, model dimensions, selected layers, seed, fitting settings, SHA-256 hashes of input files and fitted lens, raw per-prompt readouts, generated responses, and summary statistics. Fitting was checkpointed and resumable. Cross-model tables and Figure 1 were generated directly from accepted JSON outputs.

4. Results

4.1 Primary and robustness results

Table 1. Held-out primary results

Model	Role	Primary layer	Accepted affiliation tokens	Mean Delta ARC	Bootstrap 95% CI	Sign-flip p	Cohen's dz	Decision
SmolLM2-135M-Instruct	Prospectively specified primary	19/30	9	-0.142	[-0.759, 0.485]	0.6793	-0.12	Not supported
Qwen2.5-0.5B-Instruct	Exploratory robustness	15/24	13	0.720	[0.423, 1.025]	0.0018	1.27	Supported in pilot

SmolLM2 did not support H1. The estimate was close to zero relative to its uncertainty and slightly negative. Five of twelve held-out pairs produced positive differences and seven negative differences.

Qwen produced a large positive effect under the unchanged metric and prompt set. Ten of twelve held-out pairs were positive. Because Qwen was introduced after the Smol result was observed, its p-value and confidence interval describe the exploratory run and should not be treated as a second independent confirmatory test.

4.2 Depth profile

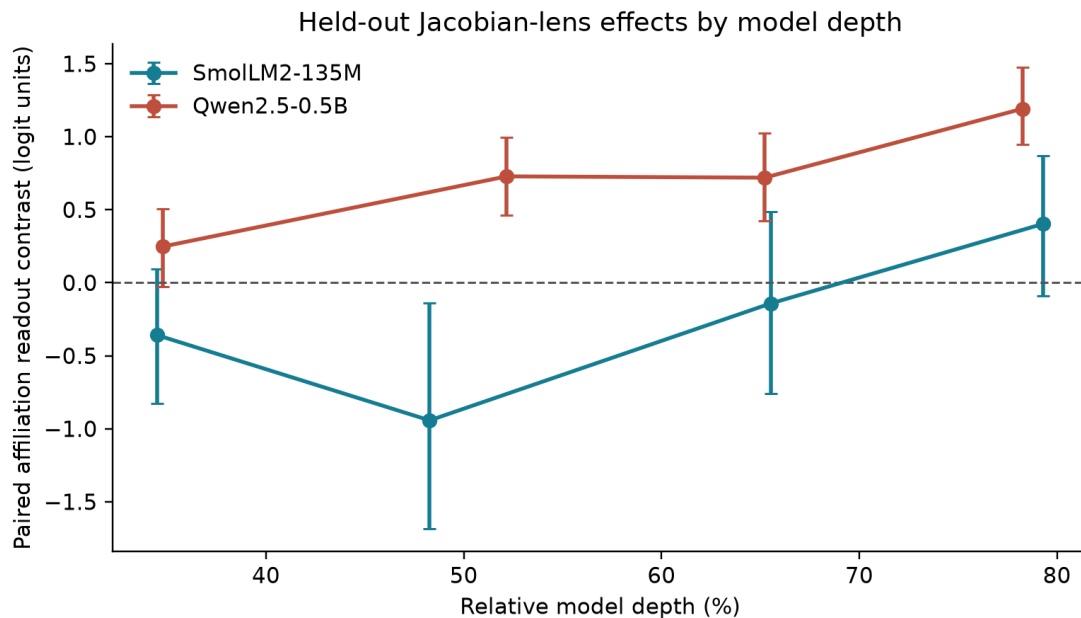


Figure 1. Held-out paired affiliation readout contrasts by relative model depth. Points are means and bars are paired bootstrap 95% intervals.

Table 2. Held-out effects across fitted layers

Model	Relative depth	Mean Delta ARC	Bootstrap 95% CI	Sign-flip p
SmolLM2	34%	-0.358	[-0.828, 0.093]	0.1762

Model	Relative depth	Mean Delta ARC	Bootstrap 95% CI	Sign-flip p
SmolLM2	48%	-0.943	[-1.684, -0.141]	0.0538
SmolLM2	66%	-0.142	[-0.759, 0.485]	0.6793
SmolLM2	79%	0.404	[-0.089, 0.872]	0.1433
Qwen2.5	35%	0.248	[-0.028, 0.504]	0.1056
Qwen2.5	52%	0.729	[0.461, 0.994]	0.0006
Qwen2.5	65%	0.720	[0.423, 1.025]	0.0018
Qwen2.5	78%	1.192	[0.947, 1.477]	0.0005

The depth profiles differed qualitatively. Qwen’s estimate increased after the earliest fitted layer and was positive with intervals above zero at the three later layers. SmolLM2 was negative through the primary layer and became positive only at the latest fitted layer, where the interval still crossed zero. Since the layer analyses beyond the designated primary layer are exploratory, they identify a pattern for replication rather than establish layer-specific effects.

4.3 Surface outputs

Explicit affiliation terms were rare in generated answers. The held-out affiliation-minus-neutral surface difference was 0.786 terms per 100 words for SmolLM2 and 0.439 for Qwen. In both cases the bootstrap interval included zero, the median paired difference was zero, and only a small minority of pairs contributed nonzero differences.

This creates a possible latent-output dissociation: Qwen showed a robust internal readout shift while rarely adding explicit affiliation markers to its short answers. The result is compatible with the claim that output-only auditing may miss internal context representations. It is not decisive evidence for that claim because the internal metric may partly reflect lexical material inherited from the prompt.

4.4 Tokenization asymmetry

SmolLM2 fragmented all ten Spanish affiliation candidates and all six Spanish controls into multiple tokens. Qwen admitted *nosotros*, *amigo*, *acuerdo*, and *primo* as single tokens, while admitting every Spanish control. The apparent cross-model difference is therefore entangled with multilingual tokenization and cannot be attributed solely to parameter count or representational organization.

5. Discussion

5.1 What the experiment establishes

First, the experiment establishes feasibility. A full Jacobian-lens fitting and paired evaluation pipeline can run on consumer hardware, preserve provenance, and produce uncertainty-aware results. This lowers the barrier to testing alignment concepts against internal representations rather than relying only on output anecdotes.

Second, the Qwen result shows that affiliation framing can be associated with a substantial shift toward affiliation-related vocabulary in a Jacobian-lens readout. The effect was not confined to one pair and appeared across multiple later layers. This is evidence for internal readout sensitivity in one model under one operationalization.

Third, the failed SmolLM2 replication is part of the finding. It rejects any simple claim that affiliation framing necessarily yields a detectable J-space signature across instruction-tuned models. The divergence

suggests boundary conditions involving capacity, language coverage, tokenization, training data, post-training, or lens quality.

The practical implication of a replicated positive result would be that relational form can matter in addition to propositional content. Two prompts requesting the same analysis could prepare measurably different internal contexts because one places the exchange inside a relation of shared identity or cooperation. Output-only auditing might then miss a change that remains latent, affects later turns, or modifies the probability of validation and agreement without producing an obvious affiliation word in the immediate answer.

That implication remains conditional in this pilot. Qwen provides evidence of internal readout sensitivity under the present operationalization, not proof that the model formed a socially meaningful bond. The SmolLM2 result shows that even this narrower sensitivity is not model-general.

5.2 What the experiment does not establish

The experiment does not show that Qwen possesses a stable affiliation module. It does not show that the measured concepts caused the generated answer. It does not show that the user would trust or obey the model more. It does not show consciousness, self-awareness, recursive intention, deception, or legal mens rea.

Most importantly, treatment prompts contained affiliation vocabulary. A Jacobian-lens readout at the final input position may retain or transform recently read lexical content. Although the substantive task was matched, the study did not include a same-token control in which affiliation words appeared in quotation, negation, or an unrelated metalinguistic instruction. The Qwen effect could therefore be lexical priming rather than recruitment of a broader affiliation state.

Nor does observational readout establish causality. To show that an affiliation-related direction participates in producing a response, a subsequent study would have to intervene on that direction, for example by ablating, substituting, or steering it, and then measure a selective change in affiliative behavior while checking that general language competence and task performance remain intact. To show the complete SPAS mechanism, a further human experiment would have to establish that matched outputs alter trust, agreement, compliance, or another reception outcome.

5.3 Implications for SPAS measurement

The original AS-Score remains valuable because it is external and architecture-independent. This pilot suggests a complementary internal metric, but not a replacement. A mature SPAS evaluation should separate at least four stages:

1. affiliation cues present in the user's input;
2. internal model representations associated with affiliation;
3. affiliative properties of the model's output;
4. behavioral effects on the human receiver.

Conflating these stages would reproduce the same inferential mistake that mechanistic interpretability is meant to reduce. The current experiment measures Stage 2 imperfectly, records a sparse proxy for Stage 3, and does not measure Stage 4.

5.4 Next decisive experiments

The next study should use lexical controls that preserve the relevant words while changing their relational function. For example, an affiliation condition could say "We are on the same team," while a control says "Quote the words 'we are on the same team' only as an example of language you must not adopt." A delayed neutral suffix can further test whether the signal persists after equivalent intervening text.

Second, the experiment should include causal interventions. Affiliation-related J-lens directions can be ablated or swapped while holding the remaining activation approximately fixed. A meaningful causal result would show a selective reduction in validation, ingroup framing, or sycophancy without a comparable decline in factual accuracy or instruction following.

Third, the fitting corpus should expand from twelve to at least one hundred generic sequences, following the reference implementation's observation that approximately one hundred prompts can already be usable. Lenses should be fitted with multiple corpus seeds to quantify estimator stability.

Fourth, the evaluation should span several model families and sizes. The present disagreement makes cross-model replication a requirement, not an optional embellishment. Multilingual claims require tokenizers and concept methods that support multi-token phrases, rather than silently discarding most Spanish concepts.

Finally, a preregistered human experiment must link internally and externally measured affiliation to trust, agreement, repair intentions, or compliance. Cheng et al. (2025) show that sycophantic responses can increase users' conviction and reduce intentions to repair interpersonal conflict. The SPAS mechanism requires evidence that affiliation contributes beyond propositional quality and overt agreement.

6. Limitations

This pilot has eight material limitations. The models are small and not representative of frontier systems. Only two models were tested. The lens-fitting corpus is much smaller than that used in the reference paper. The Qwen run was selected after the first result and is exploratory. The prompt treatment changes lexical content and length. The metric excludes multi-token concepts, producing unequal multilingual coverage. The surface measure is a sparse dictionary count rather than the full AS-Score. No causal intervention or human outcome was measured.

The uncertainty intervals account for prompt-pair sampling within this small set, not uncertainty from model selection, lens-fitting corpus choice, tokenizer choice, concept-dictionary construction, or generation settings. The p-values should therefore not be interpreted as capturing the full researcher's degrees of freedom or as supporting a universal model claim.

7. Conclusion

The Jacobian lens can make the internal side of affiliation signaling empirically accessible, but this pilot does not yield a model-general answer. SmolLM2 failed the fixed primary test. Qwen showed a large positive shift under the same operationalization, accompanied by broader token coverage and a distinct depth profile. Surface markers remained sparse.

The responsible conclusion is neither that SPAS has been mechanistically proven nor that the null model refutes it. The result is a boundary map: internal affiliation readouts are measurable, model dependence is substantial, lexical priming remains unresolved, and output-only metrics capture a different stage of the proposed process. The next advance requires same-token controls, causal intervention, larger and repeated lens fits, cross-family replication, and human reception data.

Lakatosian Epistemological Declaration

This study treats the SPAS account as part of a revisable protective belt within the Extended Phenotype Theory research programme. The hard-core claim that cultural and technological outputs can construct environments that alter selection and behavior is not tested directly here. The auxiliary claim tested is narrower: affiliation framing should generate a lens-readable internal disposition toward affiliation concepts. The SmolLM2 null result and cross-model disagreement count as genuine constraints on that auxiliary claim. In the sense of Lakatos (1978), the programme is progressive only if such constraints lead to more discriminating predictions rather than ad hoc immunization. The fixed hypothesis was exposed to

possible falsification in Popper's (1959) sense; the resulting boundary conditions remain subject to evolutionary and selectionist analysis in the traditions of Campbell (1974) and Hull (1988).

AI Disclosure

Artificial intelligence systems were used extensively as research assistants in this study. They assisted in translating the theoretical question into a computational protocol, proposing and comparing operationalizations, writing and debugging the experimental code, executing local runs under human authorization, checking statistical summaries, drafting prose, and preparing the document. The author selected the research question, approved the prospective protocol, required preservation of the negative result, determined the interpretation and limits of the findings, and accepts full responsibility for the code, analysis, claims, citations, and errors. No AI system is listed as an author. The author is an independent legal researcher and does not claim specialist training in mechanistic interpretability; for that reason, the code, raw outputs, assumptions, and failed or null results are disclosed for technical scrutiny.

Data and Code Availability

Code, prompt pairs, fitting corpus, run manifests, raw readouts, fitted-lens hashes, summaries, and the cross-model figure are available in the public repository at <https://github.com/adrianlerer/jacobian-lens>. The fork derives from Anthropic's Apache-2.0 reference implementation and preserves its attribution. Model weights are not redistributed; they remain governed by their respective Hugging Face repositories and licenses.

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